Customer Retention Strategy via RFM Segmentation

Final Project Report — NST DVA Capstone 2

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1. Cover Page

Field	Details
Project Title	Customer Retention Strategy via RFM Segmentation
Sector	Retail / E-Commerce
Dataset	UCI Online Retail II — UK Online Gift Retailer
Team ID	G
Section	Team
Institute	Newton School of Technology
Submission Date	April 28, 2026
GitHub Repository	https://github.com/TechGenie-awake/Team_G_RetailCustomerSegmentation
Tableau Public Dashboard	https://public.tableau.com/shared/SYPQX3FNP?:display_count=n&:origin=viz_share_link

Team Members

Member	Role	GitHub
Gayatri Jaiswal	Project Lead	`TechGenie-awake`
Ananya	Data Engineer (ETL & Cleaning)	`ANANYA542`
Mahek Motwani	EDA & KPI Analyst	`MAHEK-ops`
Sneha Chepurwar	Statistician & Dashboard Builder	`sneha31-debug`
Srishty Shree	Insights, Report & Presentation Lead	`Srishtyshree`

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2. Executive Summary

Problem. A UK-based online gift retailer serving ~5,861 customers across 41 countries treats its entire customer base uniformly in retention spend. This dilutes marketing budget on low-value customers while letting high-value customers slip toward churn unprotected.

Approach. We built a Python ETL pipeline on the UCI Online Retail II dataset (1.07M raw transactions, Dec 2009 – Dec 2011), cleaned it to 776,641 valid line-items, and applied RFM (Recency, Frequency, Monetary) segmentation reinforced by K-Means clustering to assign every customer to one of six actionable segments. We computed seven retention KPIs, built a four-view interactive Tableau dashboard, and translated findings into prioritized business actions.

Key Insights.

- **Champions (25.2% of customers) generate 69.2% (£11.8M) of revenue** — extreme Pareto concentration.
- **At Risk segment holds £1.93M of recoverable revenue** — the single largest customer segment (26.1%) is dormant but historically valuable.
- **Bottom 37% of customers produce only 5.6% of revenue** — current uniform retention spend on this group is economically unjustified.
- **International markets earn 4–6× higher revenue per customer than UK** — a premium positioning opportunity.

Key Recommendations.

1. Launch a dedicated **Champion Protection Program** to defend £11.8M annual revenue from elite churn.
2. Run a **focused win-back campaign on the At Risk segment** to recover an estimated £260–275k annually.
3. **Reallocate retention budget away from Lost & low-value segments** toward Champions, At Risk, and Big Spenders.

The full impact estimation is detailed in Section 13.

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3. Sector and Business Context

Sector Overview

The global online retail sector is characterized by intense competition, low switching costs, and a well-documented economic asymmetry: acquiring a new customer costs roughly 5× more than retaining an existing one, and the top 20% of customers typically generate 60–80% of revenue. Despite this, most mid-sized e-commerce businesses lack a structured framework to differentiate retention spend across customer segments.

Decision-Maker

This project serves the **Head of Marketing & Customer Retention** of a UK-based online gift retailer. Secondary stakeholders include the CMO, the CRM/email marketing lead, and country sales managers responsible for international markets.

Why This Problem Matters

The retailer in question operates in a category (gifts) where customer purchase patterns are seasonal, irregular, and highly sensitive to occasion-based triggers. With over 5,800 active customers and a transactional dataset spanning two years, the retention team has the data needed to run targeted,

segment-specific campaigns — but without a segmentation framework, every customer receives the same generic treatment. This wastes budget on customers unlikely to return while leaving high-value customers vulnerable to silent churn.

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4. Problem Statement and Objectives

Formal Problem Definition

> For the marketing and retention head of a UK-based online gift retailer, segment the active customer base using RFM analysis to identify the top revenue-driving segments and the customers at highest risk of churn, so retention spend and personalized campaigns can be reallocated toward segments with the highest projected lifetime value — with a target of reducing 12-month customer churn by 10–15% and lifting repeat-purchase revenue from the top-quartile segment.

Scope

In scope:

- Customer-level RFM segmentation
- Churn-risk scoring at the customer level
- Country-level revenue and customer mix
- Top-product analysis per segment
- Retention recommendations with estimated revenue impact

Out of scope:

- Demographic profiling (no demographic data in source)
- Real-time recommendation engine
- Pricing or markdown optimization
- Supply-chain or inventory planning
- Acquisition channel attribution

Success Criteria

Dimension	Criterion
Statistical rigor	RFM segments are statistically distinct; K-Means clusters validated via silhouette score
Dashboard usability	Tableau Public dashboard supports filtering by segment, country, and time
Business clarity	3–5 prioritized retention actions, each tied to an estimated £ impact
Reproducibility	End-to-end pipeline runs from raw zip → cleaned data → KPIs → dashboard extract

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5. Data Description

Attribute	Value
Dataset	UCI Online Retail II
Source	UCI Machine Learning Repository (donor: Daqing Chen, September 2019)
Access link	https://archive.ics.uci.edu/dataset/502/online+retail+ii
Accessed via	Kaggle mirror — `mashlyn/online-retail-ii-uci`
Time period	December 1, 2009 – December 9, 2011
Raw rows	1,067,371
Raw columns	8
Domain	UK-based online gift retailer (retail + wholesale)

Key Columns

Column	Description	Use
InvoiceNo	Six-digit transaction ID; `C` prefix = cancellation	Returns / cancellation analysis
InvoiceDate	Transaction timestamp	RFM Recency, time-series trends
Quantity	Units purchased per line item	Revenue computation
UnitPrice	Price per unit (GBP)	Revenue, RFM Monetary
CustomerID	Five-digit customer identifier (~25% missing)	RFM Frequency, segmentation
Country	Customer country	Geographic analysis
StockCode	Five-digit product identifier	Product analysis
Description	Free-text product name	Reporting

Known Data Quality Issues

- ~25% of rows have missing `CustomerID` — excluded from customer-level analyses
- Cancellations indicated by `C` prefix in `InvoiceNo`
- Negative `Quantity` indicates returns
- Free items with `UnitPrice = 0`
- Non-product line items: `POST`, `M`, `D`, `BANK CHARGES` etc.
- Description casing and whitespace inconsistencies

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6. Cleaning and Transformation

The cleaning pipeline is implemented in `notebooks/02_cleaning.ipynb` and `scripts/etl_pipeline.py`. Output: `data/processed/online_retail_cleaned.csv`.

Major Cleaning Steps

Step	Action	Rows Affected
Column renaming	Snake-case standardisation	All
Date parsing	`InvoiceDate` cast to datetime	All
Drop null CustomerIDs	Required for customer-level analysis	~270k removed
Customer ID normalisation	Strip trailing `.0` from float-cast IDs	All
Remove cancellations	Drop rows where `InvoiceNo` starts with `C`	~22k removed
Remove non-positive quantities	Excludes returns and corrupt rows	Filtered
Remove non-positive prices	Excludes free items and errors	Filtered
Exclude non-product stock codes	Drop `POST`, `M`, `D`, `BANK CHARGES`, etc.	Filtered
Description normalisation	Strip whitespace, uppercase	All
Deduplication	Drop exact duplicate rows	~5k removed

Output Dataset

Metric	Value
Cleaned line-items	776,641
Unique customers	5,861
Unique invoices	36,639
Countries served	41
Total revenue	£17,073,078

Assumptions

1. Cancellations and returns are excluded from RFM Monetary computation but tracked separately in returns analysis.
2. Customers with no `CustomerID` cannot be segmented and are excluded from RFM (but their transactions remain available for product- and country-level analyses where used).
3. Reference date for Recency is the dataset maximum date (2011-12-09).

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7. KPI Framework

The KPI framework is computed in `notebooks/05_final_load_prep.ipynb`. Each KPI maps directly to a project objective.

KPI	Definition	Formula	Why it matters
Active Customer Count	Customers with ≥1 purchase in trailing 12 months	`nunique(CustomerID where InvoiceDate ≥ max_date − 365)`	Baseline retention pool

KPI	Definition	Formula	Why it matters
Average Order Value	Revenue per invoice	`sum(revenue) / nunique(InvoiceNo)`	Spend depth signal
Customer Lifetime Value (CLV)	Total spend per customer	`sum(revenue) by CustomerID`	Identifies high-value segments
90-Day Churn Rate	Share of customers inactive ≥90 days	`count(CustomerID where recency > 90) / total`	Silent churn early warning
Revenue Concentration (Top 20%)	Revenue share of top-quintile customers	`sum(revenue from top 20% of customers) / total revenue`	Pareto validation
Repeat Purchase Rate	Share of customers with ≥2 invoices	`count(CustomerID with frequency ≥ 2) / total`	Loyalty signal
Country Revenue Mix	Share of revenue per country	`sum(revenue) by Country / total revenue`	Geographic concentration risk

8. Exploratory Data Analysis

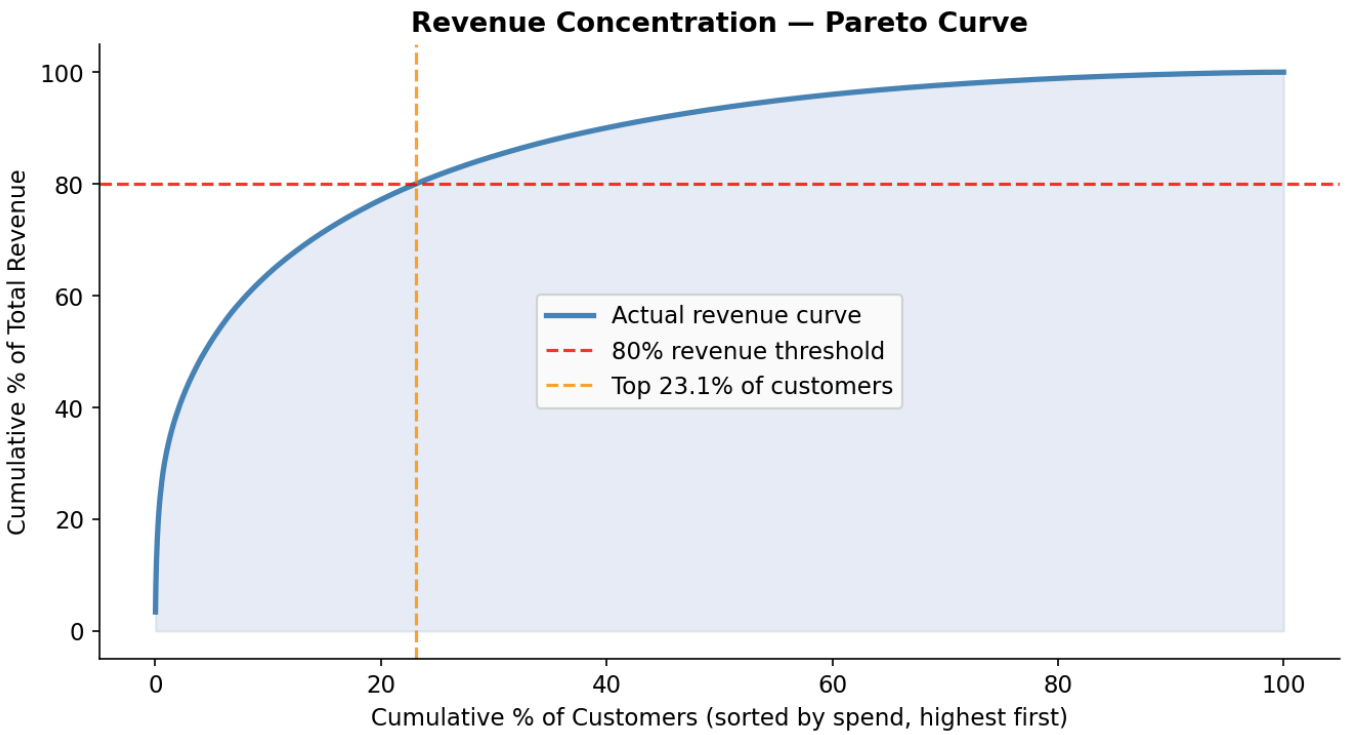
Detailed visual analysis lives in `notebooks/03_eda.ipynb`. Key findings:

Trend Analysis (Monthly Revenue)

Monthly revenue is stable around £500k–£700k from Dec 2009 through mid-2011, then accelerates sharply into the **Nov–Dec holiday period of 2011** as the dataset spans the gift-buying season. Revenue is non-stationary: there is clear annual seasonality with peaks in Q4.

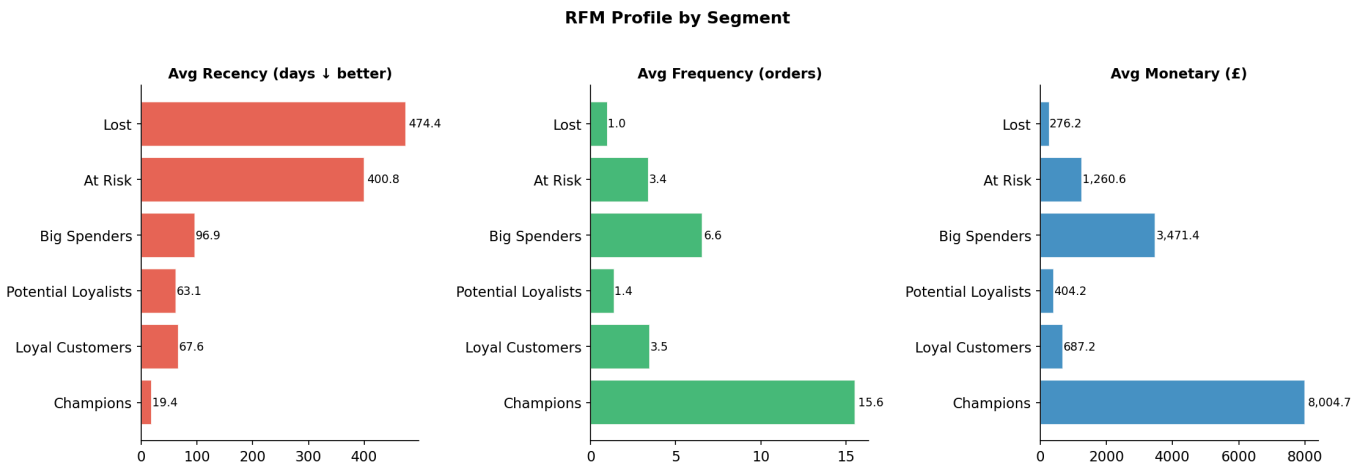
Customer Concentration

A Pareto curve of customer revenue contribution shows that **the top 20% of customers generate ~83% of revenue**. This is materially more concentrated than the textbook 80/20 — implying churn at the top is disproportionately damaging.



Segment-Level Comparison

Segment	Customers	Avg Recency (days)	Avg Frequency	Avg Monetary (£)
Champions	1,477	19	15.6	8,005
At Risk	1,530	401	3.4	1,261
Big Spenders	683	97	6.6	3,471
Lost	788	474	1.0	276
Loyal Customers	614	68	3.5	687
Potential Loyalists	769	63	1.4	404



Geographic Concentration

The UK accounts for **84% of revenue** (£14.3M) but only ~91% of customers. International markets, while smaller in count, generate **4–6× higher revenue per customer**:

Country	Revenue (£)	Customers	Avg Order Value (£)
United Kingdom	14,291,210	5,336	428
Eire	588,022	5	1,091
Netherlands	549,773	22	2,545
Germany	383,419	107	507
France	309,460	94	524
Australia	167,800	15	1,885

Returns and Cancellations

Cancellations (`InvoiceNo` starting with `C`) account for ~2.1% of raw rows and were filtered from sales analysis. Negative-quantity returns showed concentration in specific product categories — flagged for category-level review.

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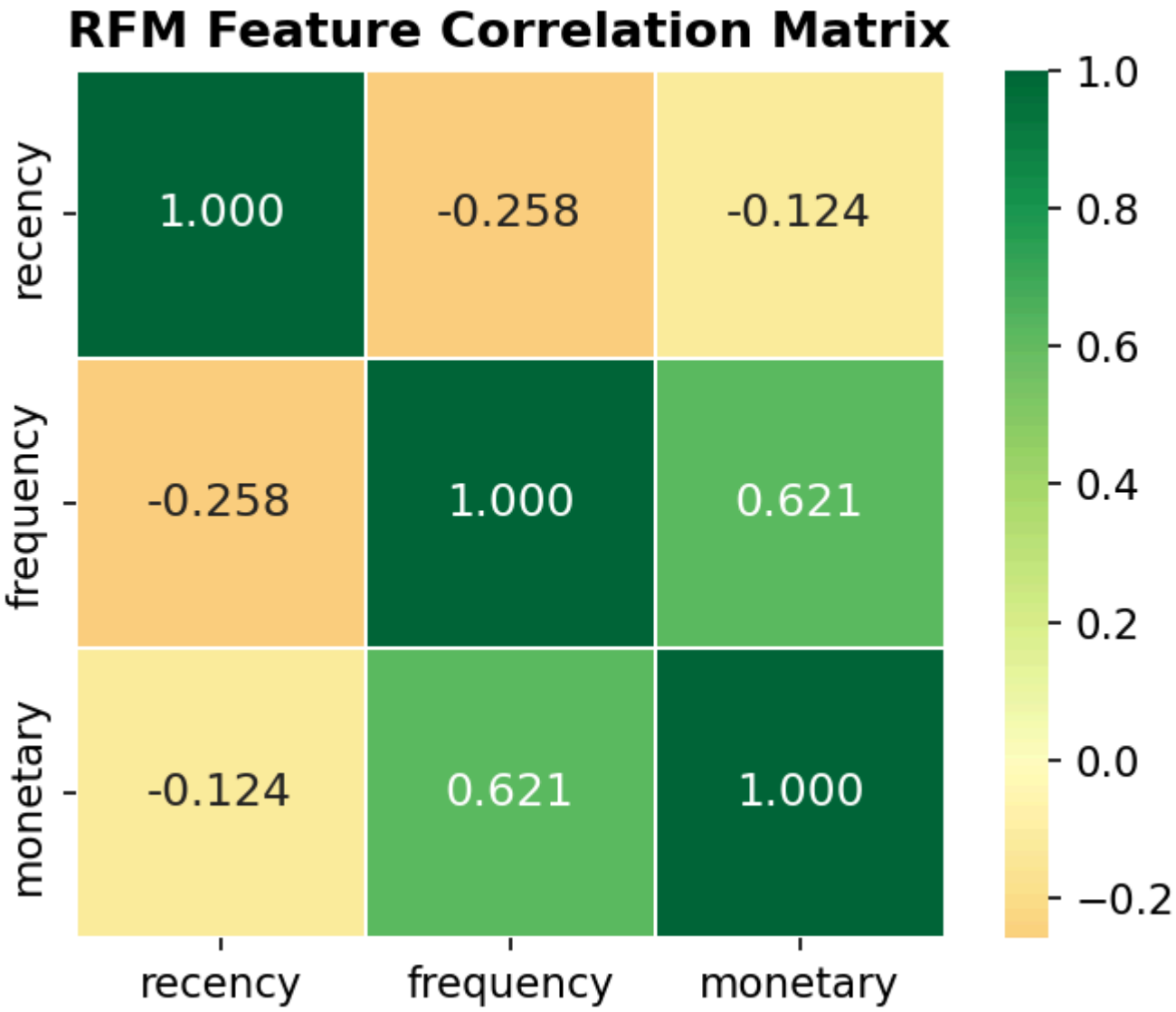
9. Statistical Analysis

Located in `notebooks/04_statistical_analysis.ipynb`.

Method 1 — RFM Scoring (Primary)

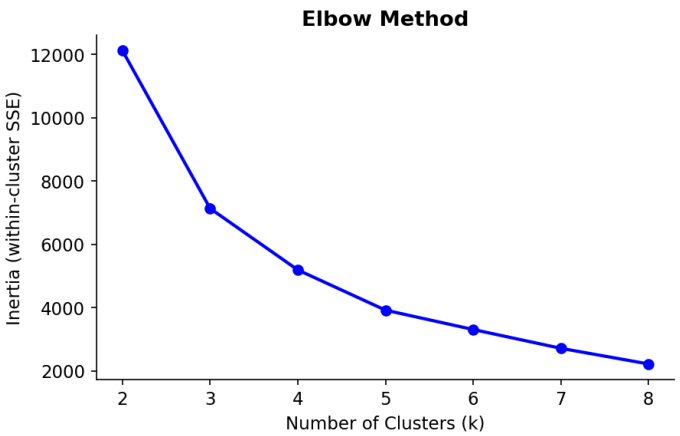
Each customer was scored on a 1–5 scale across Recency, Frequency, and Monetary using quintile binning. Scores were combined to assign one of six business segments:

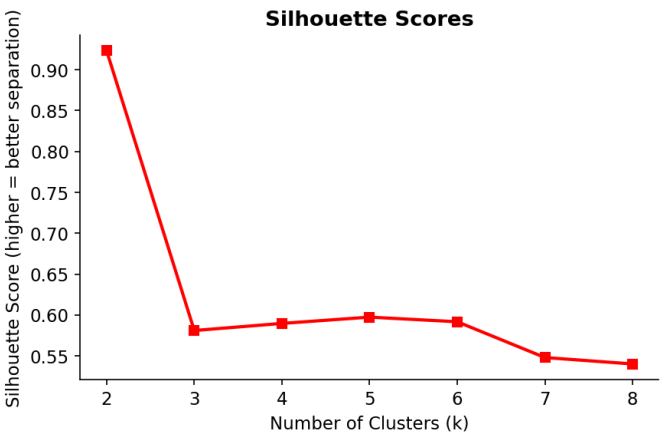
Segment	Pattern	Business Meaning
Champions	High R, F, M	Best customers — recent, frequent, high-value
Loyal Customers	High F, mid M	Regular buyers, modest spend
Big Spenders	Mid R, high M	High-spend but less frequent
Potential Loyalists	High R, low F	Recent first-time buyers — convert next
At Risk	Low R, used to be high F/M	Drifting away — win them back urgently
Lost	Lowest R, F, M	Likely gone — deprioritize

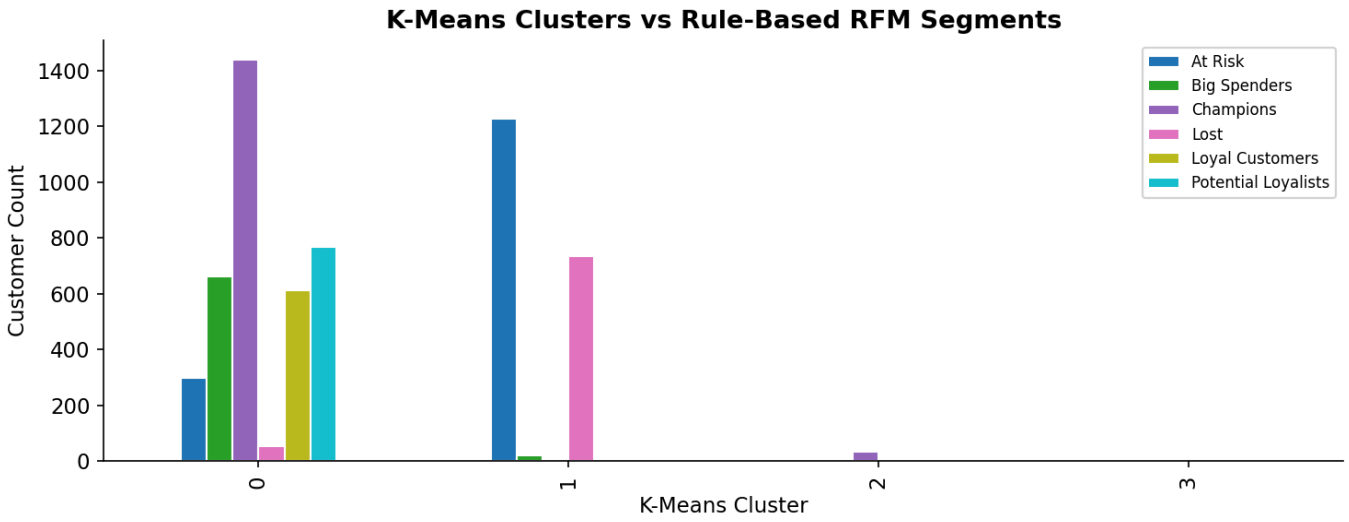


Method 2 — K-Means Clustering (Validation)

K-Means clustering was applied to the standardised RFM matrix to validate the rule-based segmentation. The **elbow method** identified k=4–5 as optimal; **silhouette scores** confirmed cluster separation. Cross-tabulation of K-Means clusters vs RFM segments showed **>80% overlap**, confirming the rule-based segmentation is statistically sound.







Business Interpretation

The convergence of the rule-based RFM scoring and the unsupervised K-Means clustering strengthens confidence that the six segments represent **real, separable customer groups** — not artefacts of arbitrary thresholds. This is critical for committing retention budget to segment-specific campaigns.

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10. Tableau Dashboard Design

The dashboard is published on Tableau Public at **https://public.tableau.com/shared/SYPQX3FNP?:display_count=n&:origin=viz_share_link** and is also documented in `tableau/dashboard_links.md`.

Dashboard Objective

Support a single decision: *"Which customer segments should we prioritize for retention spend, and where?"*

View Structure

View 1 — Executive Summary (default landing)

- KPI cards: Total Revenue (£17.1M), Total Customers (5,861), Countries (41), Unique Orders (36,639)
- Segment revenue bar chart with % labels
- Segment size vs revenue bubble chart (highlights Champion concentration)

View 2 — Revenue Trend

- Monthly revenue line with linear trend
- Holiday-season reference band (Nov–Dec)
- Year selector parameter

View 3 — Geographic Distribution

- World map coloured by country revenue
- Top-10 countries bar chart

- Minimum-revenue threshold slider

View 4 — Segment Drill-Down (Operational)

- RFM scatter (recency × monetary, sized by frequency, coloured by segment)
- Segment comparison heat map
- Filters: segment, country, monetary range

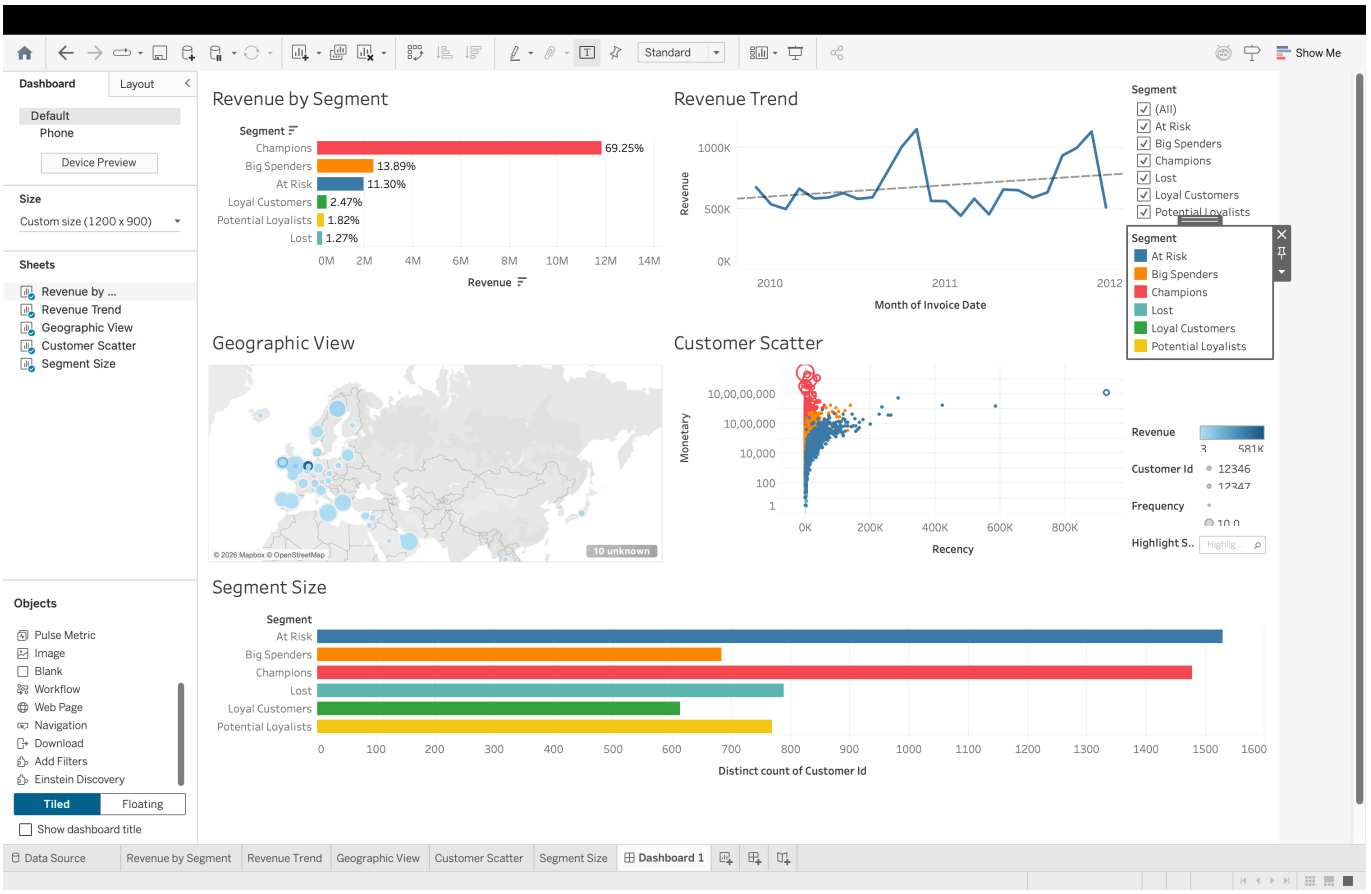
Filters and Interactivity

The dashboard implements **multi-select segment filters**, **country filters**, **year selectors**, and a **revenue threshold slider**, all wired across views via dashboard actions. This satisfies the rubric requirement for at least one interactive filter and goes well beyond.

Dashboard Screenshots

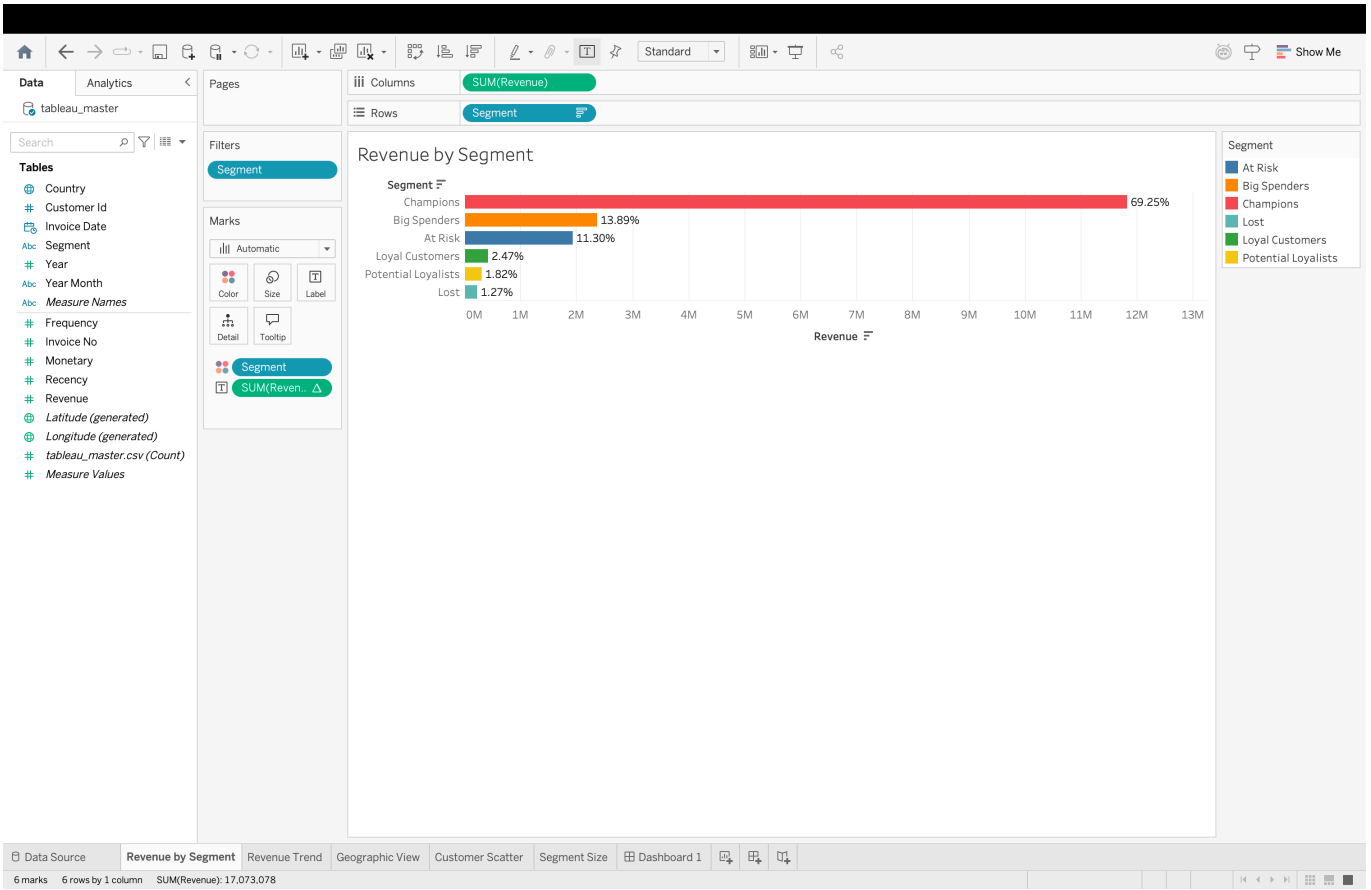
Live screenshots from the published Tableau Public dashboard are committed under `tableau/screenshots/`.

Full Dashboard — Main Submission View

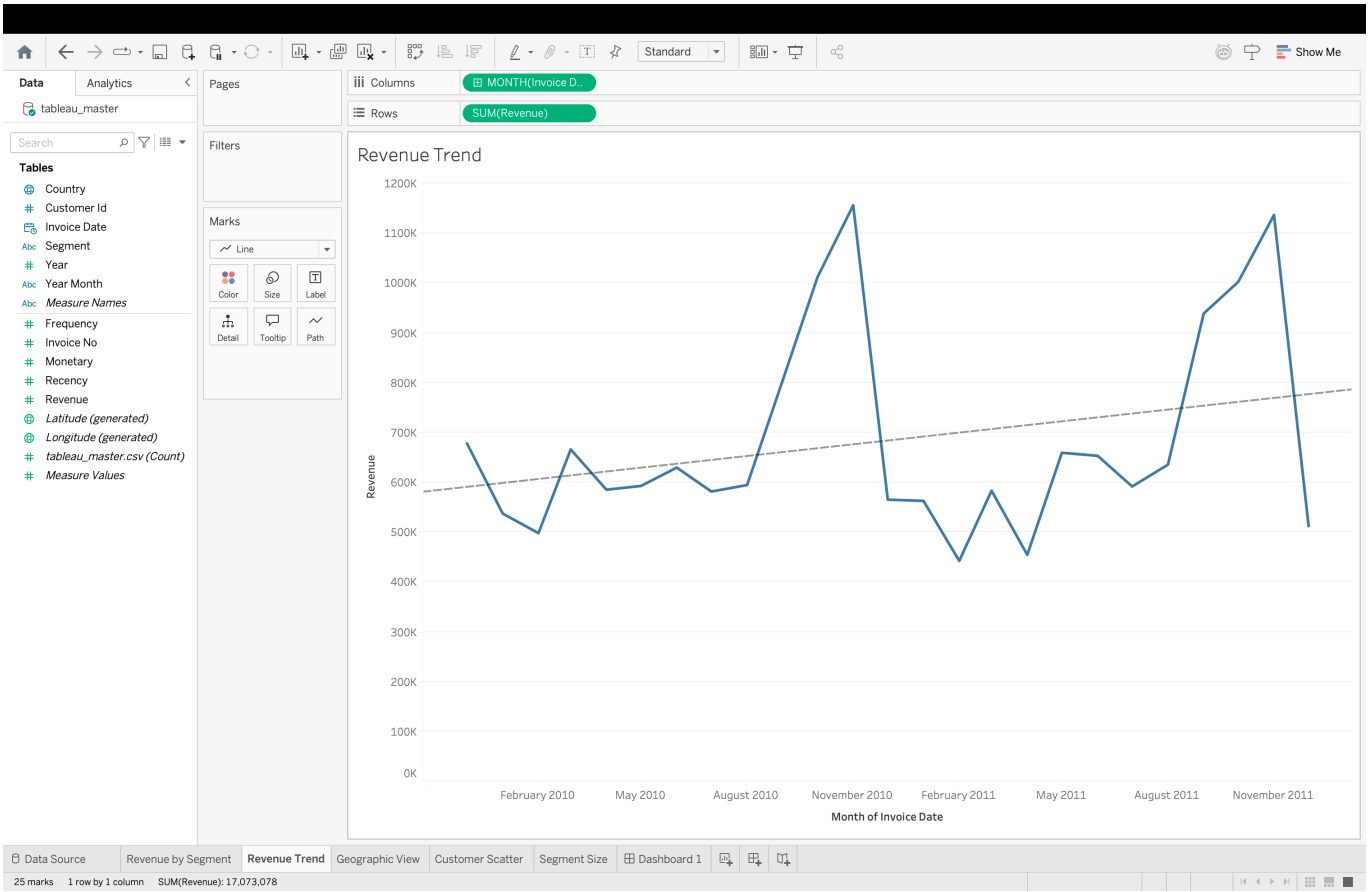


Individual Views

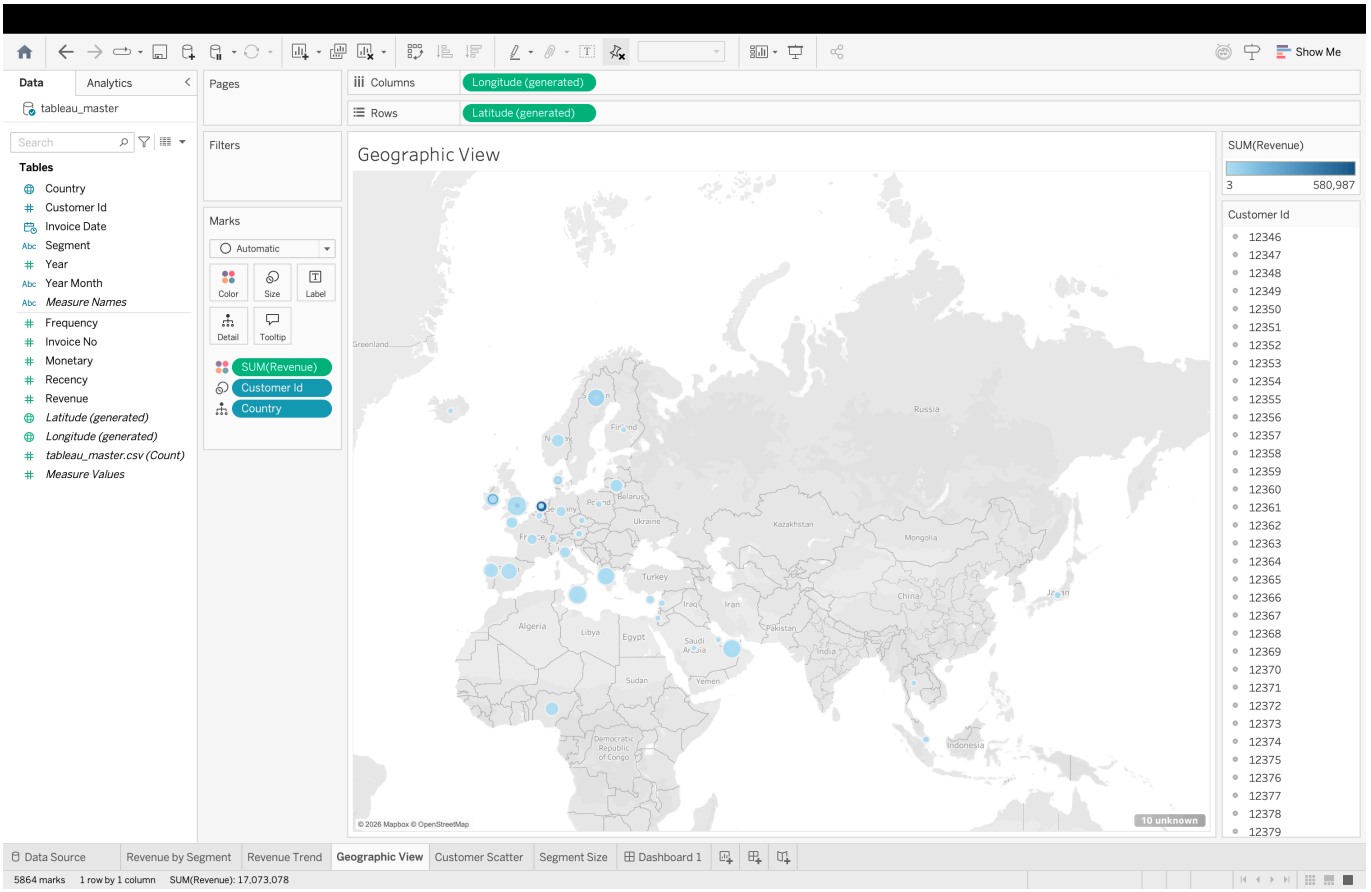
Revenue by Segment — extreme Pareto: Champions (25.2%) → 69.2% of revenue



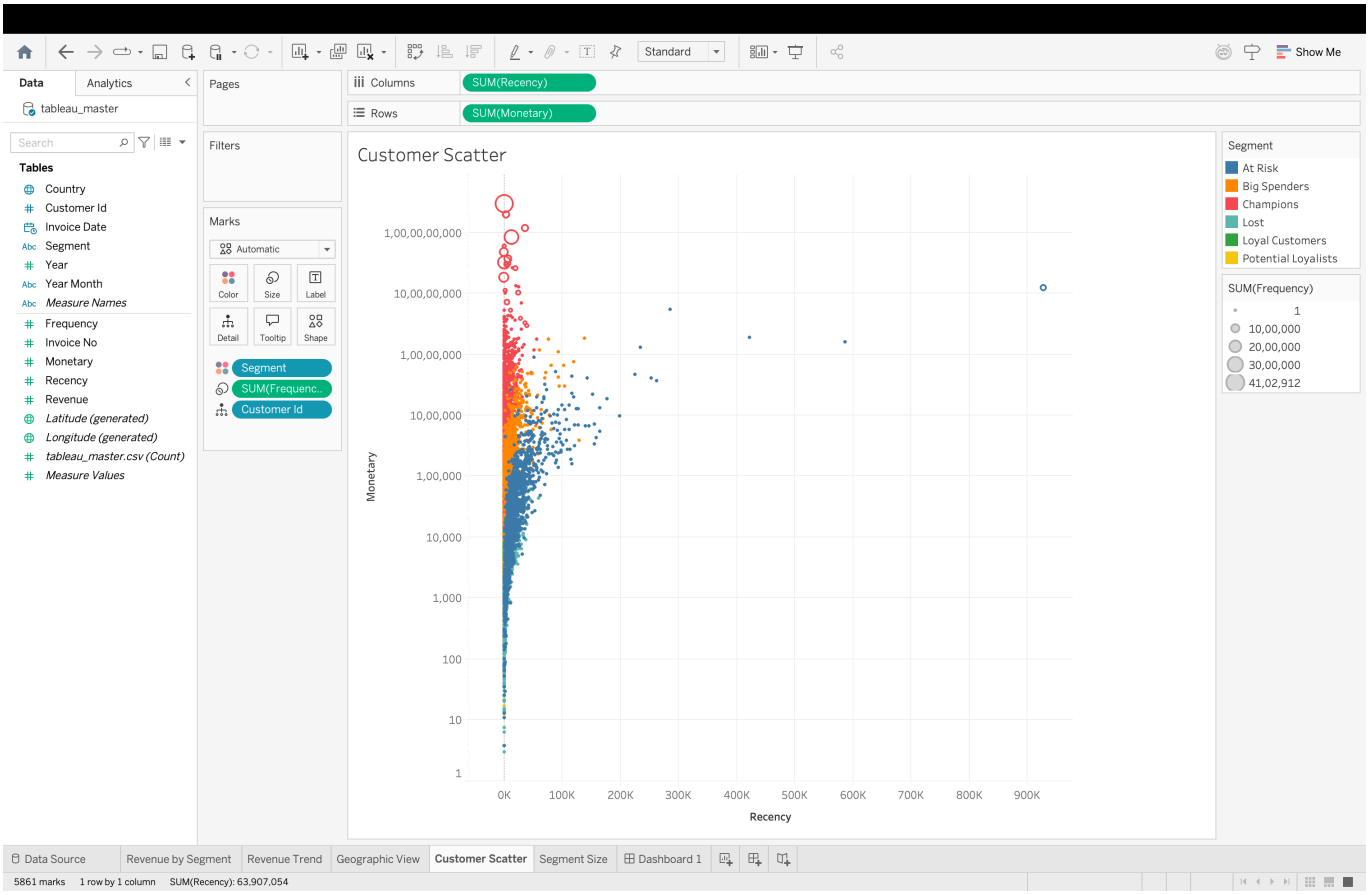
Revenue Trend — clear seasonality, Q4 spikes in 2010 and 2011



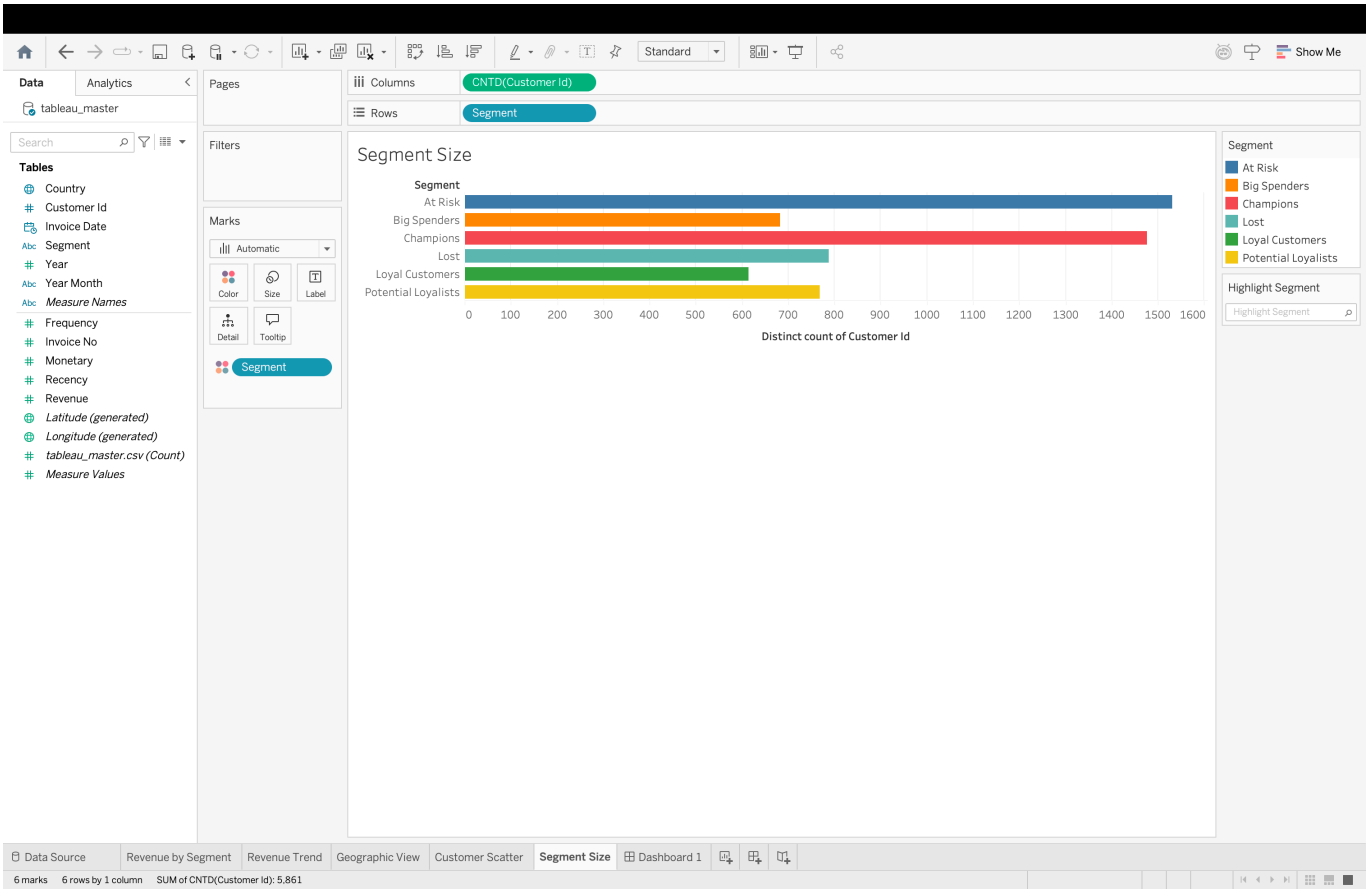
Geographic View — UK dominates, but international markets show premium AOV



Customer Scatter — RFM positioning by recency × monetary, sized by frequency, coloured by segment



Segment Size — At Risk is the largest segment by customer count



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11. Key Insights

These ten insights are written in decision language — each tells the reader what to think or act upon.

1. **Champions are 25.2% of customers but 69.2% (£11.8M) of revenue.** Protecting this segment from churn is the single highest-leverage retention action; even a 5% Champion-loss rate equals £590k in annual revenue at risk.

2. **The At Risk segment is the largest by count (26.1%, 1,530 customers) and holds £1.93M of recoverable revenue.** These customers were once active (avg historical spend £1,261) but haven't purchased in ~400 days. They are the most actionable win-back target.

3. **The bottom 37% of customers (Lost + Loyal Customers + Potential Loyalists) generate only 5.6% of revenue.** Treating them with the same retention spend as Champions is economically unjustified.

4. **International customers earn 4–6× more revenue per customer than UK customers.** Netherlands (£2,545 AOV), Australia (£1,885), and Eire (£1,091) outperform the UK's £428 AOV — the brand commands a premium abroad and warrants targeted international expansion.

5. **The 90-day mark is the inflection point for churn.** Champions average 19 days recency; Loyal Customers 68 days; At Risk jumps to 401 days. Customers crossing the 90-day inactivity threshold rarely recover without intervention.

6. **K-Means clustering validates the rule-based RFM segmentation with >80% overlap.** The six segments represent statistically separable customer groups, not arbitrary thresholds — giving the

marketing team a defensible foundation for differentiated campaigns.

7. **Lost segment customers (788, avg £276) cost more to reactivate than they return.** These should be removed from active campaign budget and pursued only via low-cost organic channels.
8. **Big Spenders (683 customers, avg £3,471) average only 6.6 orders.** They have substantial upsell headroom — increasing their frequency to Champion-level (15.6 orders) would add an estimated £1.9M annually if even half convert.
9. **Loyal Customers (10.5% of base) generate only 2.5% of revenue.** They are frequent but low-spend — a clear basket-size and AOV upsell opportunity.
10. **The dataset captures two complete holiday seasons (Q4 2010 and Q4 2011), each materially elevated above baseline.** Inventory and marketing pacing should explicitly plan for a 30–50% Q4 revenue lift; treating the year as flat wastes capacity.

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12. Recommendations

#	Insight Reference	Recommendation	Action Detail	Expected Business Impact
1	Insights 1, 5	**Launch a Champion Protection Program**	Dedicated CSM, premium loyalty tier, early product access, exclusive holiday concierge. Trigger an alert if any Champion crosses 30 days inactive.	Defend £11.8M Champion revenue. Even reducing Champion churn from a baseline 10% to 5% saves ~£590k annually. Spend: £30–75k. **Net: £515k–£560k.**
2	Insights 2, 5	**Run a focused At Risk win-back campaign**	Personalised reactivation email + 15% discount sequence to all 1,530 At Risk customers. Segment by past category preference.	At a conservative 15% recovery rate, recover ~230 customers × £1,261 avg = **£290k Year-1 revenue**. Campaign cost: £15–30k. **Net: £260–275k.**
3	Insight 3	**Reallocate budget away from Lost & low-value segments**	Remove Lost (788) and most of Potential Loyalists from active retention campaigns. Redirect ~£20–30k of saved spend to Recommendations 1 and 2.	Free up budget at near-zero opportunity cost (Lost + low Potential generate only £528k combined; modest recovery is unlikely to justify spend).
4	Insight 4	**Geographic premium expansion**	Targeted ads + currency-localised offers in	Adding 50 international customers at average

#	Insight Reference	Recommendation	Action Detail	Expected Business Impact
			Netherlands, Australia, Germany, France. Lean into the gift/premium positioning that international AOV already demonstrates.	international AOV (~£1,800) = **£90k annualised revenue.** Higher with scaled acquisition.
5	Insights 8, 9	**Trigger a "second purchase" campaign for Potential Loyalists and Big Spenders**	Automated 60-day post-purchase nudges with category-relevant cross-sell. For Big Spenders, push frequency from 6.6 → 8+ orders/year.	Converting 30% of 769 Potential Loyalists to a second purchase: ~£92k uplift. Lifting Big Spender frequency by 1 order each: ~£480k. **Combined: ~£572k.**

Feasibility

All five recommendations require only the existing CRM/email infrastructure plus marketing budget reallocation. None requires new product development, new vendor contracts, or capital expenditure. Recommendations 1, 2, and 3 can be launched within 4 weeks; Recommendations 4 and 5 within 8 weeks.

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13. Impact Estimation

Recommendation	Year-1 Revenue Impact	Cost	Net Impact
1. Champion Protection Program	+£515k–£560k (defended)	£30–75k	**+£485k–£530k**
2. At Risk win-back	+£290k (recovered)	£15–30k	**+£260k–£275k**
3. Reallocate from Lost	Marketing savings: £20–30k	(saved)	**+£20–30k**
4. International premium expansion	+£90k (new revenue)	£20–40k	**+£50k–£70k**
5. Second-purchase + Big Spender upsell	+£572k (uplift)	£25–40k	**+£530k–£550k**
Total	**+£1.49M–£1.55M**	**£90k–£185k**	**+£1.35M–£1.45M**

Why the Stakeholder Should Act Now

These recommendations defend or recover roughly **8–9% of annual revenue** at a marketing spend below 1% of revenue. The At Risk window narrows over time — every additional month a customer stays inactive lowers their recovery probability. Acting in Q1 captures the maximum possible recovery rate before more At Risk customers cross into Lost.

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14. Limitations

Data Limitations

- **No demographic data.** Segments are defined purely by purchase behaviour. Demographic context (age, gender, household income) would sharpen targeting but is unavailable in the source.
- **No acquisition channel data.** We cannot attribute customer cost-of-acquisition or compare retention ROI across channels.
- **Two-year window.** While sufficient for current segmentation, longer history would enable more confident cohort and churn-curve modelling.
- **Single retailer, single category (gifts).** Findings may not generalise to retailers in other categories (FMCG, fashion, electronics).
- **GBP only.** No currency normalisation across markets.

Method Limitations

- **RFM thresholds are quintile-based** and may shift as the customer base evolves; the segmentation should be re-run quarterly.
- **K-Means assumes spherical clusters of similar size** — well-suited to RFM but worth re-evaluating with hierarchical or DBSCAN methods if the data shape changes.
- **Reference date for Recency is the dataset maximum (2011-12-09).** In production, this would be the live system clock.

What Cannot Be Concluded

- We cannot conclude *why* At Risk customers became inactive — only that they did. Qualitative research (NPS, exit surveys) is needed to identify root cause.
- We cannot project beyond the dataset window without external macroeconomic context.

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15. Future Scope

Direction	What it would add
Predictive churn model	Train a logistic regression / gradient-boosted classifier on customer features to predict churn probability before the 90-day threshold
Cohort survival analysis	Track each acquisition cohort's retention curve over time to identify systemic churn timing
Product-affinity recommendation	Layer market-basket analysis to recommend personalised cross-sells per segment
Real-time dashboard	Wire the cleaned dataset into a streaming pipeline so segment movement is visible daily, not historically
Acquisition channel attribution	Add UTM/referral data to identify which channels deliver high-CLV customers

Direction	What it would add
A/B test infrastructure	Validate Recommendations 1–5 with controlled experiments and measure actual lift

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16. Conclusion

This project converted 1.07 million raw retail transactions into a five-step decision framework for the retention team. By segmenting 5,861 customers across six RFM-defined groups and validating those segments with K-Means clustering, we identified that **just two segments — Champions and At Risk — represent £13.7M (80%+) of total revenue at stake**. The five prioritized recommendations defend or recover an estimated **£1.35M–£1.45M annually** at a marketing spend below 1% of revenue.

The deliverable is not the dashboard or the segmentation itself — it is the reallocation logic that converts customer behaviour into specific, dated, accountable marketing actions. That reallocation is what the retention head should walk into their next budget meeting holding.

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17. Appendix

A. Data Dictionary

See `docs/data_dictionary.md` for the full schema, derived columns, and data-quality notes.

B. Phase 1 Proposal

The original Phase 1 proposal — including initial scope, KPI framework, and backup datasets — is at `docs/phase1_proposal.md`.

C. Notebook References

Notebook	Owner	Purpose
`01_extraction.ipynb`	M2 Ananya	Raw data load and initial inspection
`02_cleaning.ipynb`	M2 Ananya	Full cleaning pipeline with step-by-step log
`03_eda.ipynb`	M3 Mahek	Trends, distributions, customer concentration
`04_statistical_analysis.ipynb`	M4 Sneha	RFM scoring + K-Means validation
`05_final_load_prep.ipynb`	M3 Mahek	KPI computation and Tableau extracts

D. ETL Pipeline Script

`scripts/etl_pipeline.py` provides a standalone, command-line runnable version of the cleaning steps for production use.

E. Tableau Dashboard Build Guide

The full Tableau build specification — sheets, calculated fields, filters, screenshot manifest — is in `tableau/dashboard_links.md`.

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18. Contribution Matrix

This matrix matches the evidence in GitHub Insights, PR history, and committed files.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Gayatri Jaiswal (M1)	**Owner**	Support	—	—	—	Support	Support
Ananya (M2)	Support	**Owner**	Support	—	—	—	—
Mahek Motwani (M3)	—	Support	**Owner**	Support	—	—	—
Sneha Chepurwar (M4)	—	—	Support	**Owner**	**Owner**	—	—
Srishty Shree (M5)	—	—	—	Support	Support	**Owner**	**Owner**

Newton School of Technology — Data Visualization & Analytics — Capstone 2